SOFTWARE REQUIREMENTS SPECIFICATION

Assets Tracking System

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document describes the functional and non-functional requirements for the Assets Tracking System. This document is intended for the development team, project supervisor Dr. Salwa Osama, and all stakeholders of the SE2 project at Helwan University. This is version 2.0, updated to reflect revised role responsibilities, corrected workflows, and system fixes.

1.2 Scope

The Assets Tracking System is a web-based application that enables organizations to manage their physical assets throughout the entire asset lifecycle — from creation and assignment to maintenance and retirement. The system supports three user roles: Admin, Asset Manager, and Employee, each with clearly defined permissions.

1.3 Definitions & Acronyms

Term	Definition
Admin	Super-user with full system access including user management and system configuration
Asset Manager	Operational role responsible for asset lifecycle, assignments, and maintenance tickets
Employee	End user who receives assigned assets and reports issues to Asset Manager
AVAILABLE	Asset status: asset is free and ready to be assigned
ASSIGNED	Asset status: asset is currently assigned to an employee
UNDER_MAINTENANCE	Asset status: asset has an open or in-progress maintenance ticket
RETIRED	Asset status: asset is permanently decommissioned — terminal state
SRS	Software Requirements Specification
AOP	Aspect-Oriented Programming — used for transparent audit logging with timestamp
OCL	Object Constraint Language — formal constraints on the class model

1.4 Overview

This document is organized as follows:

- Section 2 — Overall System Description
- Section 3 — System Features & Functional Requirements
- Section 4 — Role Permissions Matrix
- Section 5 — Non-Functional Requirements
- Section 6 — OCL Constraints Summary
- From Section 7 — Diagrams

2. Overall System Description

2.1 Product Perspective

The Assets Tracking System is a standalone web application. It integrates AOP-based audit logging (with full timestamps) for all state-changing operations, and sends real-time notifications to relevant users on assignment, maintenance events, and system alerts.

2.2 User Classes

Role	Access Level	Primary Responsibilities
Admin	Full system access	User management, system configuration, asset category management, admin dashboard with all assets view, audit log review
Asset Manager	Operational access	Full asset lifecycle management, assignment with approval workflow, bulk CSV import, maintenance ticket management (open/update/resolve/close), notifications
Employee	Read + limited write	View assigned assets ("My Assets" dashboard), report issues to Asset Manager, receive notifications, edit own profile via Settings

2.3 Constraints

- Assets cost field must be zero (0) at creation — cost tracking is not in scope for this version.
- Asset status 'ASSIGNED' is not selectable when adding a new asset — status is set to AVAILABLE automatically on creation.
- Bulk CSV import is only available on the frontend through the Asset Manager interface.
- Admin cannot create or manage maintenance tickets — this is exclusively an Asset Manager responsibility.
- Audit log is only visible to Admin — the Asset Manager UI does not show the audit log.

3. System Features & Functional Requirements

3.1 Authentication & User Management

3.1.1 User Registration & Login

- The system shall allow users to register with a unique username and email.
- The system shall authenticate users via email and password.
- Passwords shall be stored as hashed values.
- Failed login attempts shall display a descriptive error message.

3.1.2 User Profile Settings

User Settings (Profile Edit)

- Every user (Admin, Asset Manager, Employee) shall have a Settings page to edit their own profile including name, email, and password.
- Email changes must be validated for uniqueness before saving.
- Password change requires confirmation of the old password.

3.1.3 User Deactivation

Deactivate User

- Admin can deactivate any user account.
- A deactivated user cannot log in.
- Deactivation must first return all active asset assignments belonging to that user before completing.
- If the user has open maintenance tickets assigned to them, those tickets must be re-assigned before deactivation is allowed.
- The system shall display a confirmation dialog listing the user's active assignments before proceeding with deactivation.

3.2 Asset Management

3.2.1 Create Asset — Single

- Asset Manager can create a single asset via a form with fields: name, serial number, category, location, warranty expiry, and description.
- **Status field restriction:** The 'ASSIGNED' status option must be removed from the asset creation form. New assets are automatically set to AVAILABLE upon creation.
- **Cost field:** The cost field must default to 0 and shall not be editable in this version.
- Serial numbers must be globally unique across all assets.

3.2.2 Create Asset — Bulk CSV Import

Bulk CSV Import

- Asset Manager can import multiple assets at once via CSV file upload from the frontend interface.
- The CSV template must be downloadable from the UI.

- The system shall validate each row before saving — invalid rows return an error message with the row number and reason.
- Valid rows are saved even if some rows fail validation.
- All imported assets default to status AVAILABLE and cost 0.

3.2.3 Edit Asset

Asset Editing (Fix)

- Asset Manager can edit an existing asset's name, location, category, description, and warranty expiry.
- Serial number is immutable once created.
- Status cannot be manually edited — it is controlled by the assignment and maintenance workflows.

3.2.4 Delete Asset

Asset Deletion (Fix)

- Asset Manager can delete an asset only when its status is AVAILABLE or RETIRED.
- Deletion is blocked if the asset has any OPEN or IN_PROGRESS maintenance tickets.
- The system shall display a confirmation dialog before permanently deleting an asset.

3.2.5 Admin Dashboard — All Assets

Admin: All Assets View

- The Admin dashboard shall display a complete list of all assets in the system with their current status, assigned user, and category.
- Admin can filter and search assets but cannot edit or delete them — editing is reserved for Asset Manager.

3.2.6 Employee Dashboard — My Assets

Employee: My Assets View

- The Employee dashboard shall show only assets currently assigned to the logged-in employee.
- Each asset entry shows: name, serial number, category, location, assignment date, and status.
- Employee cannot edit or return assets — they can only report an issue to the Asset Manager.

3.3 Asset Assignment

3.3.1 Create Assignment with Approval

Approval Workflow

- Asset Manager initiates an assignment by selecting an available asset and a target employee.
- The assignment is submitted as a pending request and routed to Admin for approval.
- Admin reviews the pending assignment and either approves or rejects it.

- On approval: the asset status changes to ASSIGNED, the assignment record is created, an AOP audit log entry with timestamp is recorded, and the employee receives a notification.
- On rejection: the Asset Manager receives a rejection notification with a reason, and the asset remains AVAILABLE.
- An asset can only have one active assignment at a time.
- Only active employees can be assigned assets.

3.3.2 Return Asset

- Asset Manager can return an assigned asset.
- On return: the asset status changes to AVAILABLE, the assignment record is closed with a returnedAt timestamp, and an audit log entry is recorded.

3.4 Maintenance Ticket Management

3.4.1 Ticket Roles — Clarification

Who can open tickets?

- Employees do NOT open maintenance tickets directly.
- When an employee experiences an issue with an asset, they report the issue to the Asset Manager (via notification or internal message).
- The Asset Manager reviews the report and, if warranted, opens a maintenance ticket on behalf of the employee.
- Admin cannot create, update, or manage maintenance tickets. This is exclusively the Asset Manager's domain.

3.4.2 Create Ticket

- Asset Manager creates a ticket by selecting: the asset, issue title, description, and priority (LOW / MEDIUM / HIGH).
- On creation: asset status changes to UNDER_MAINTENANCE, ticket status is set to OPEN, and an AOP audit log entry with timestamp is recorded.
- An asset that is RETIRED cannot have a ticket opened against it.

3.4.3 Ticket Lifecycle

Status Transition	Allowed By	Effect
OPEN → IN_PROGRESS	Asset Manager	Work begins on the asset
IN_PROGRESS → RESOLVED	Asset Manager	Resolution notes added; resolvedAt timestamp set
RESOLVED → CLOSED	Asset Manager	Ticket closed; asset status → AVAILABLE; audit log recorded

- No other status transitions are permitted.
- Each status change generates an AOP audit log entry with timestamp.
- Asset Manager can add notes to any non-CLOSED ticket at any time.

3.5 Notifications

Notification System (Revised)

Event	Recipient	Type
Assignment approved	Employee	**ASSIGNMENT**
Assignment rejected	Asset Manager	**ASSIGNMENT**
Asset returned	Employee	**ASSIGNMENT**
Maintenance ticket opened	Asset Manager (confirmation)	**MAINTENANCE**
Ticket status updated	Asset Manager	**MAINTENANCE**
User deactivated	Affected user (if contact exists)	**SYSTEM**
Bulk import completed	Asset Manager	**SYSTEM**

- Notifications can be marked as read.
- Unread notifications are highlighted in the UI header.

3.6 Audit Log

Audit Log (Revised Access & Timestamp)

- Every state-changing operation (asset create/edit/delete, assignment, return, ticket open/update/close, user activate/deactivate) generates an audit log entry automatically via AOP.
- **Each audit entry includes:** action, entityType, entityId, oldValue, newValue, actor (user), and a full timestamp (date + time).
- **Access restriction:** Only Admin can view the audit log. The Asset Manager interface does NOT include an audit log view.
- Audit log entries are immutable — they cannot be edited or deleted.
- Admin can filter audit logs by date range, action type, and actor.

4. Role Permissions Matrix

The following table summarizes what each role can do.

Feature / Action	Admin	Asset Manager	Employee
View all assets (dashboard)	✅	✅	❌
View own assets (My Assets)	❌	❌	✅
Create asset (single)	❌	✅	❌
Bulk CSV import	❌	✅ (frontend)	❌
Edit asset	❌	✅	❌
Delete asset	❌	✅	❌
Assign asset (initiate)	❌	✅	❌
Approve / reject assignment	✅	❌	❌
Return asset	❌	✅	❌
Open maintenance ticket	❌	✅	❌
Update ticket status	❌	✅	❌
Add ticket notes	❌	✅	❌
Report issue to Asset Manager	❌	❌	✅
Manage users (create/deactivate)	✅	❌	❌
View audit log	✅	❌	❌
Manage asset categories	✅	✅	❌

Edit own profile (Settings)	✅	✅	✅
Receive notifications	✅	✅	✅

5. Non-Functional Requirements

5.1 Security

- All passwords are stored using bcrypt hashing (minimum cost factor 10).
- Role-based access control (RBAC) is enforced on every API endpoint.
- JWT tokens are used for session management with expiry and refresh logic.
- All audit log entries are append-only and protected from modification.

5.2 Performance

- Asset listing pages shall load within 2 seconds for up to 10,000 assets.
- Bulk CSV import shall process up to 500 rows within 10 seconds.
- Notification delivery shall occur within 5 seconds of the triggering event.

5.3 Usability

- The system shall provide clear error messages for all validation failures.
- The system shall use confirmation dialogs before irreversible actions (delete, deactivate, retire).
- Notifications shall be accessible from a persistent header icon across all pages.

5.4 Maintainability

- The system shall use AOP for audit logging so that business logic is not polluted with logging code.
- The codebase shall follow clean architecture principles (separation of controller, service, and repository layers).

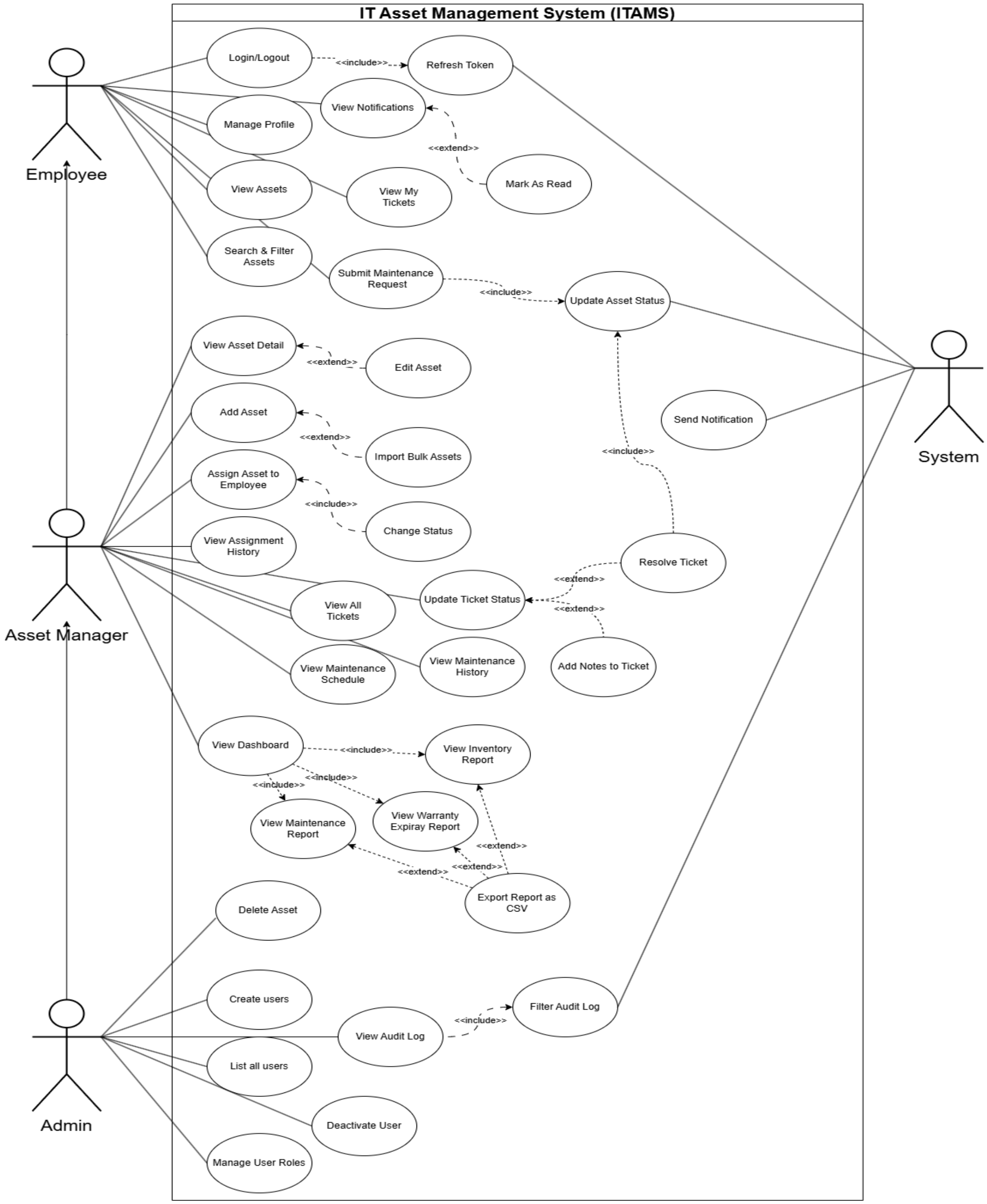
6. OCL Constraints Summary

The following OCL invariants are the most critical business rules enforced on the system model. Full OCL specification is provided in the companion document Final_OCL_Constraints.pdf.

Context	Invariant	Description
Asset	*MaxOneActiveAssignment*	An asset cannot have more than one active assignment at any time
Asset	*AssignedHasActiveAssignment*	If status=ASSIGNED, exactly one active AssetAssignment must exist
Asset	*AvailableHasNoActiveAssignment*	If status=AVAILABLE, no active assignments exist
Asset	*RetiredIsTerminal*	A RETIRED asset cannot change status or be assigned
Asset	*WarrantyAfterPurchase*	warrantyExpiry must be after purchaseDate
AssetAssignment	*ActiveNoReturn*	An active assignment has returnedAt = null
AssetAssignment	*InactiveHasReturn*	A closed assignment must have returnedAt set
MaintenanceTicket	*AssetUnderMaintenanceWhenOpen*	If ticket is OPEN or IN_PROGRESS, asset must be UNDER_MAINTENANCE
MaintenanceTicket	*ClosedHasResolvedAt*	A CLOSED ticket must have resolvedAt set
MaintenanceTicket	*ValidTransition*	Only OPEN→IN_PROGRESS→RESOLVED→CLOSED transitions allowed
User	*InactiveNoActiveAssignments*	A deactivated user must have no active asset assignments
User	*UniqueEmail*	Email must be unique across all users

7. UseCase Diagram

https://drive.google.com/file/d/1veHcWYHO9tmVc0w7M92Tv7TXF8CfSQ4b/view?usp=sharing



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8. Activity Diagrams

8.1) Assignment workflow:
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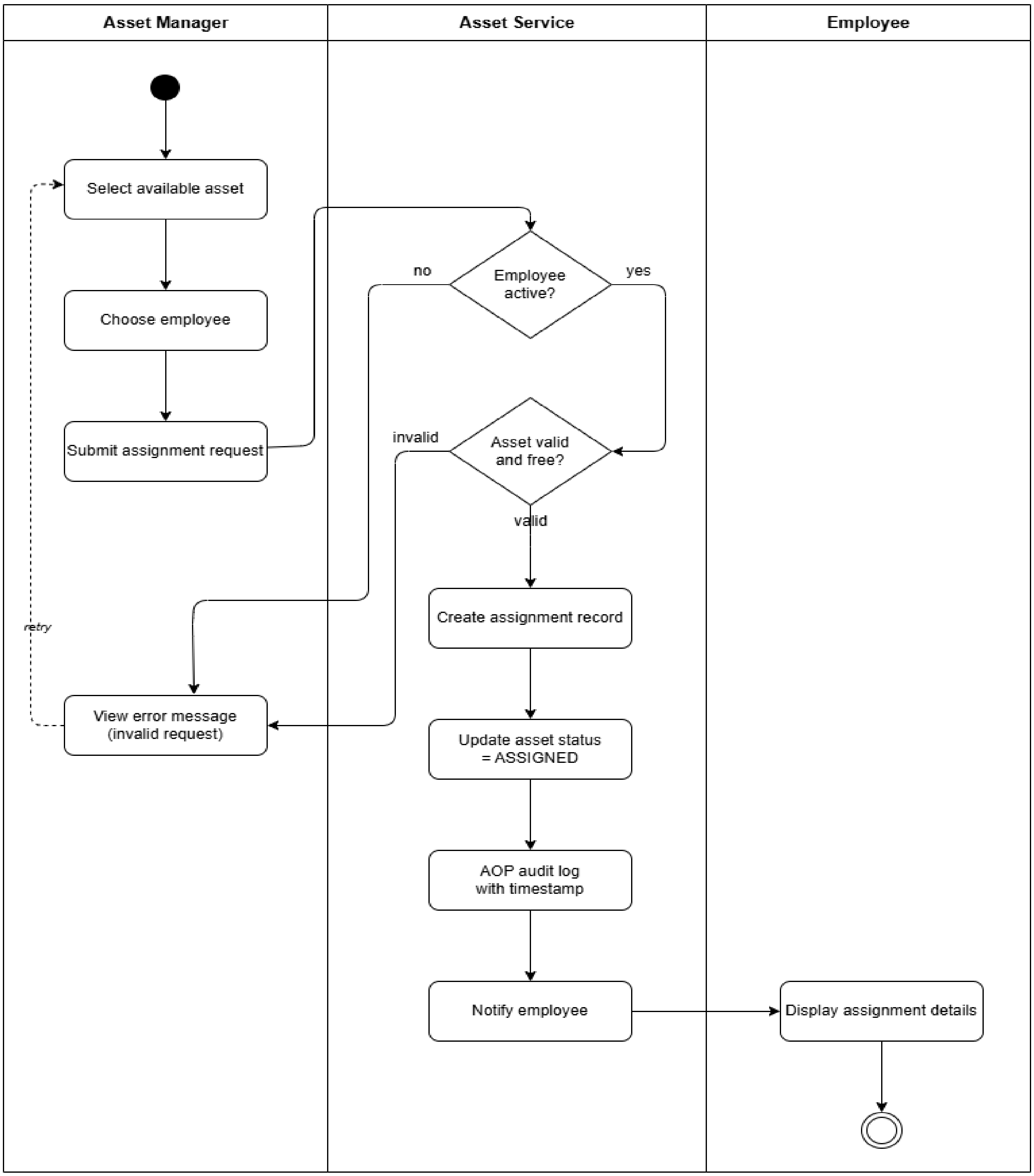
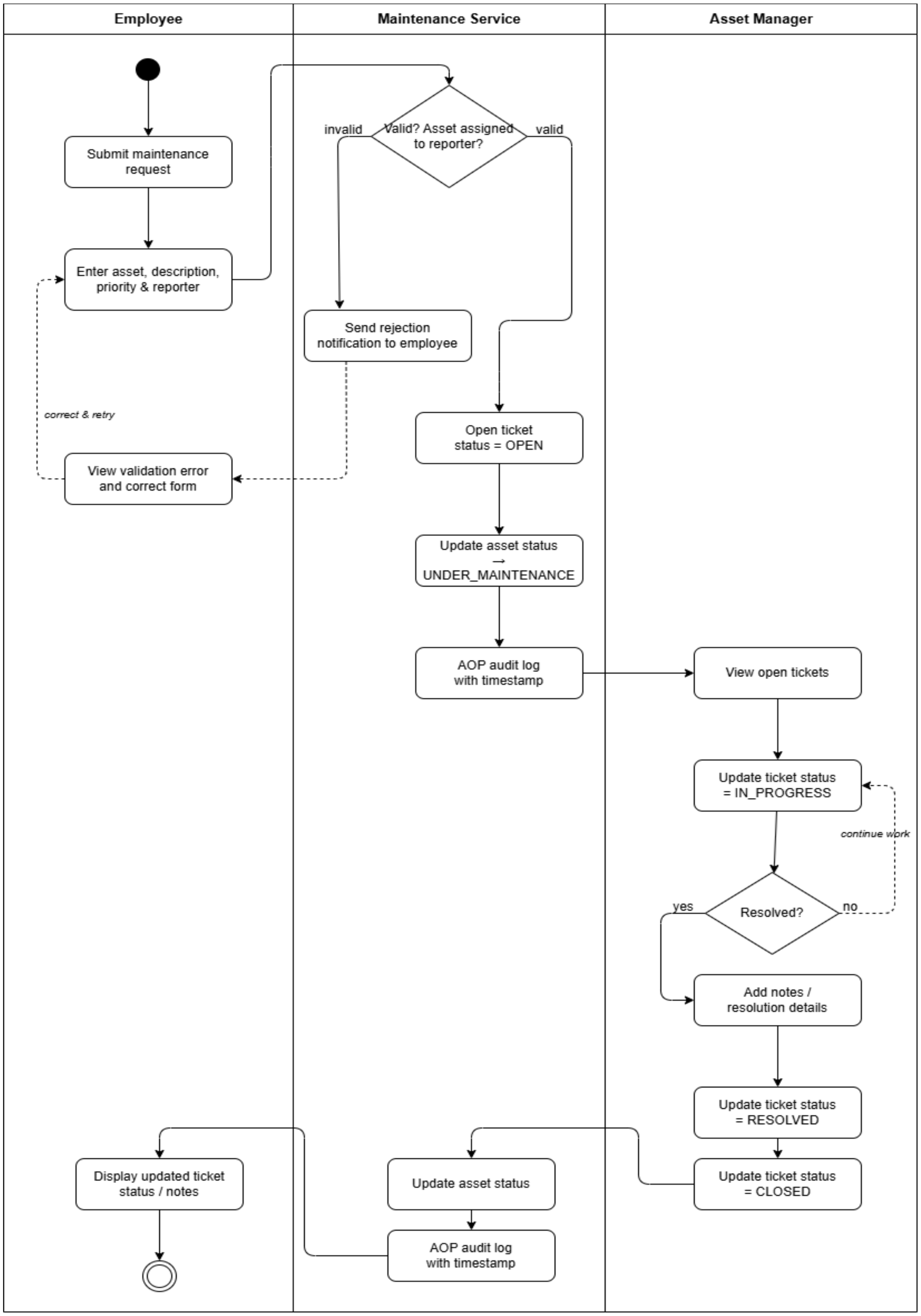
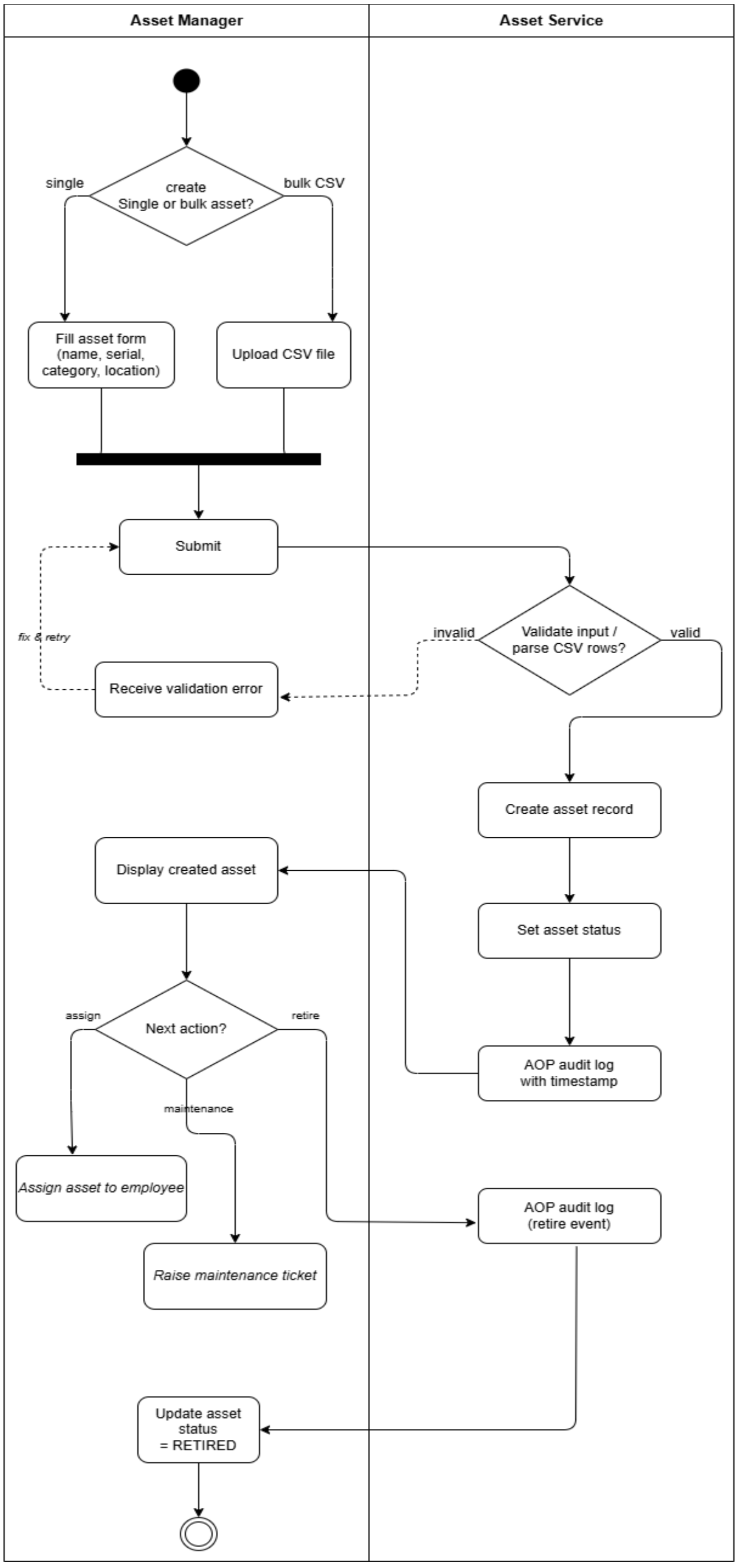


8.2) Maintenance ticket :
https://drive.google.com/file/d/1f7Vlj_Ay7Rpnl7PYTROOmolfQWR4L-rA/view?usp=sharing



8.3) Asset LifeCycle:
https://drive.google.com/file/d/18BQJah5Ka30JrasKR6Jj_dNQnngwZiNr/view?usp=sharing



8.4) User Authentication :

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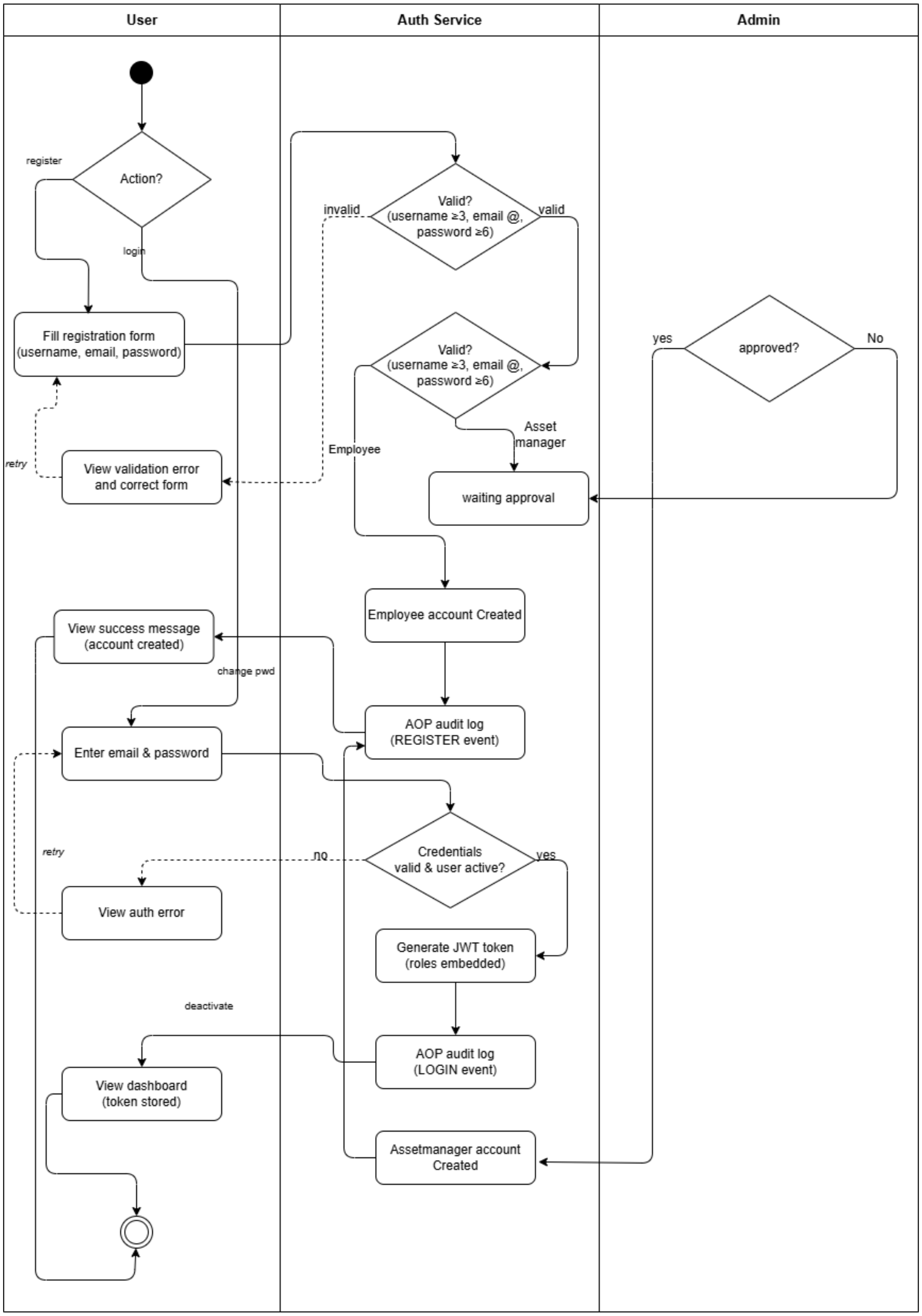
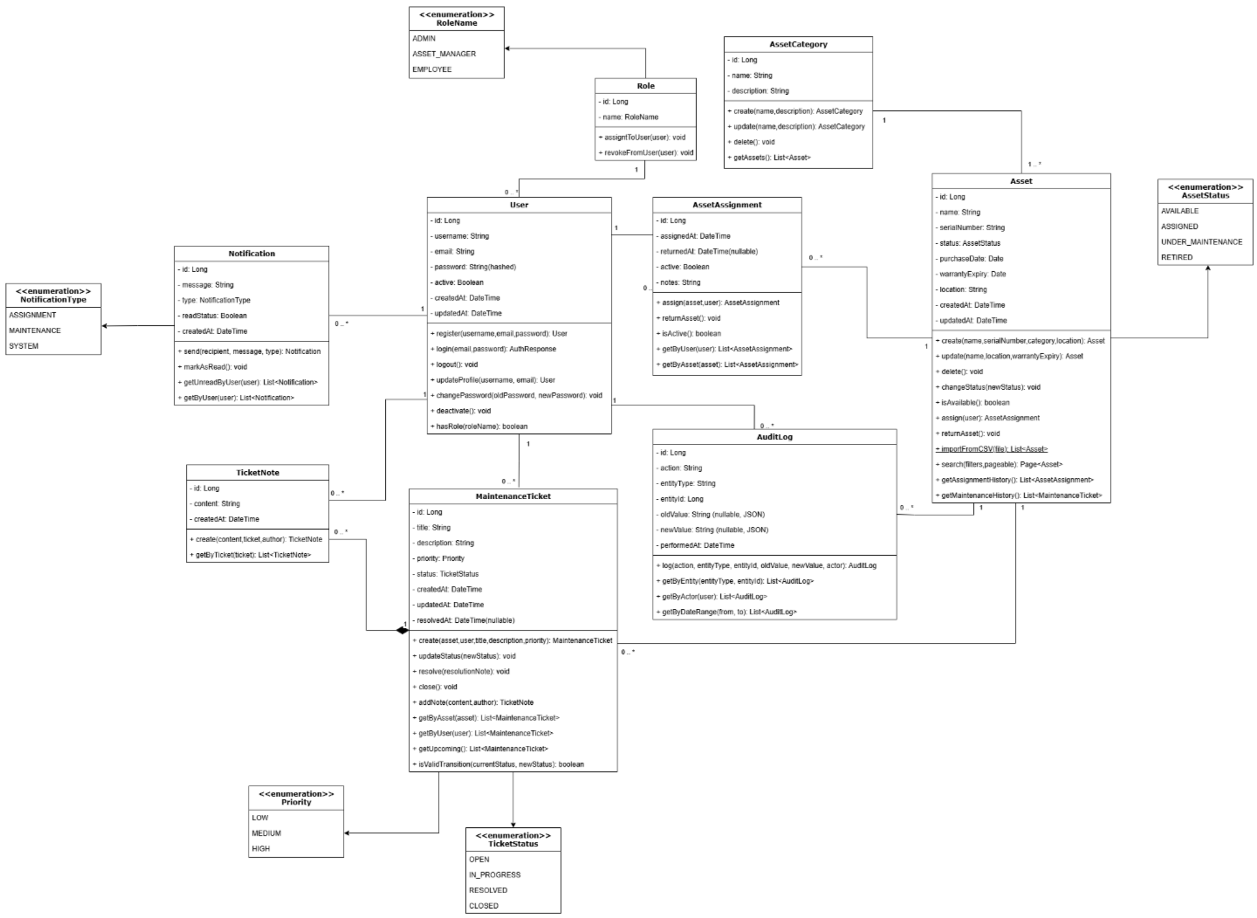


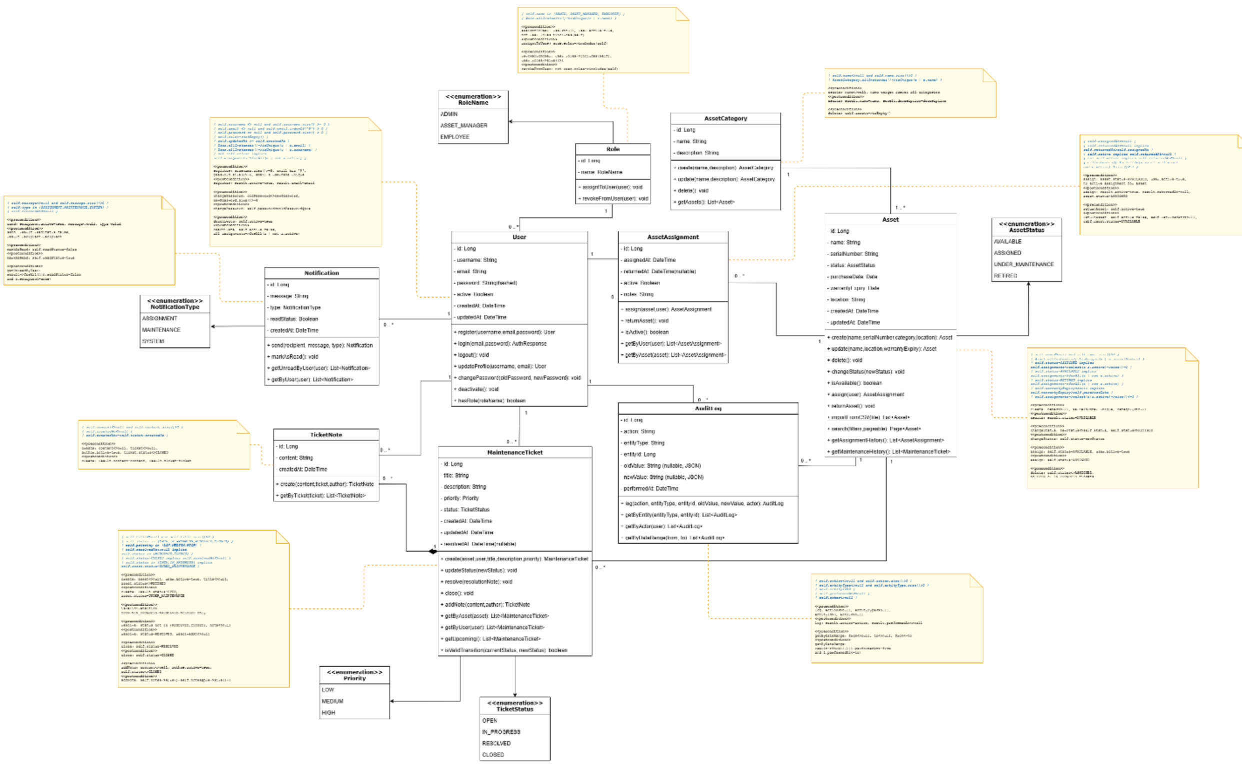
9. Class diagram

https://drive.google.com/file/d/1jhhporwZJ2BIi6dBCzat_rKMpT2j8_9o/view?usp=sharing



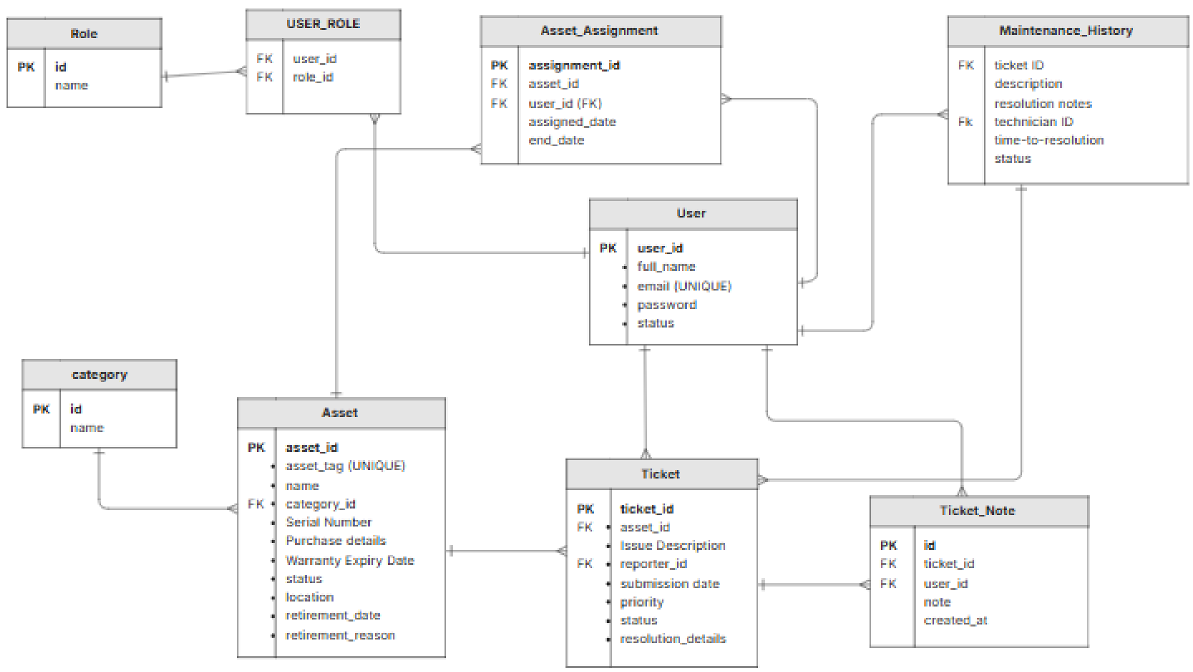
10. class Diagram with OCL

https://drive.google.com/file/d/15gru3wIOpZW8-8412LS730ESBvVxb04H/view?usp=sharing



11. ERD

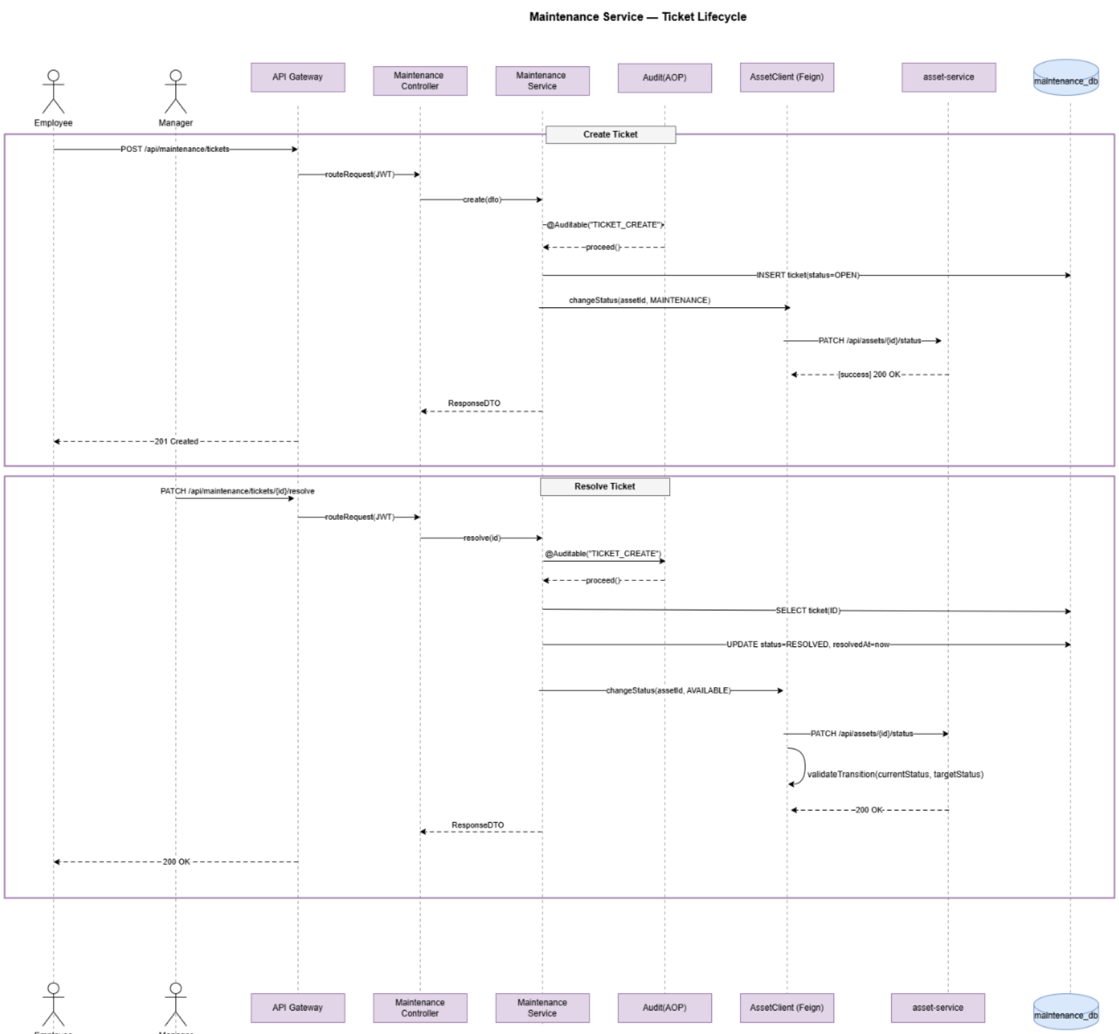
https://app.moqups.com/JYiY09Vh6Aw09uRSnK7103egI0adcNRw/view/page/ad4e74bdc



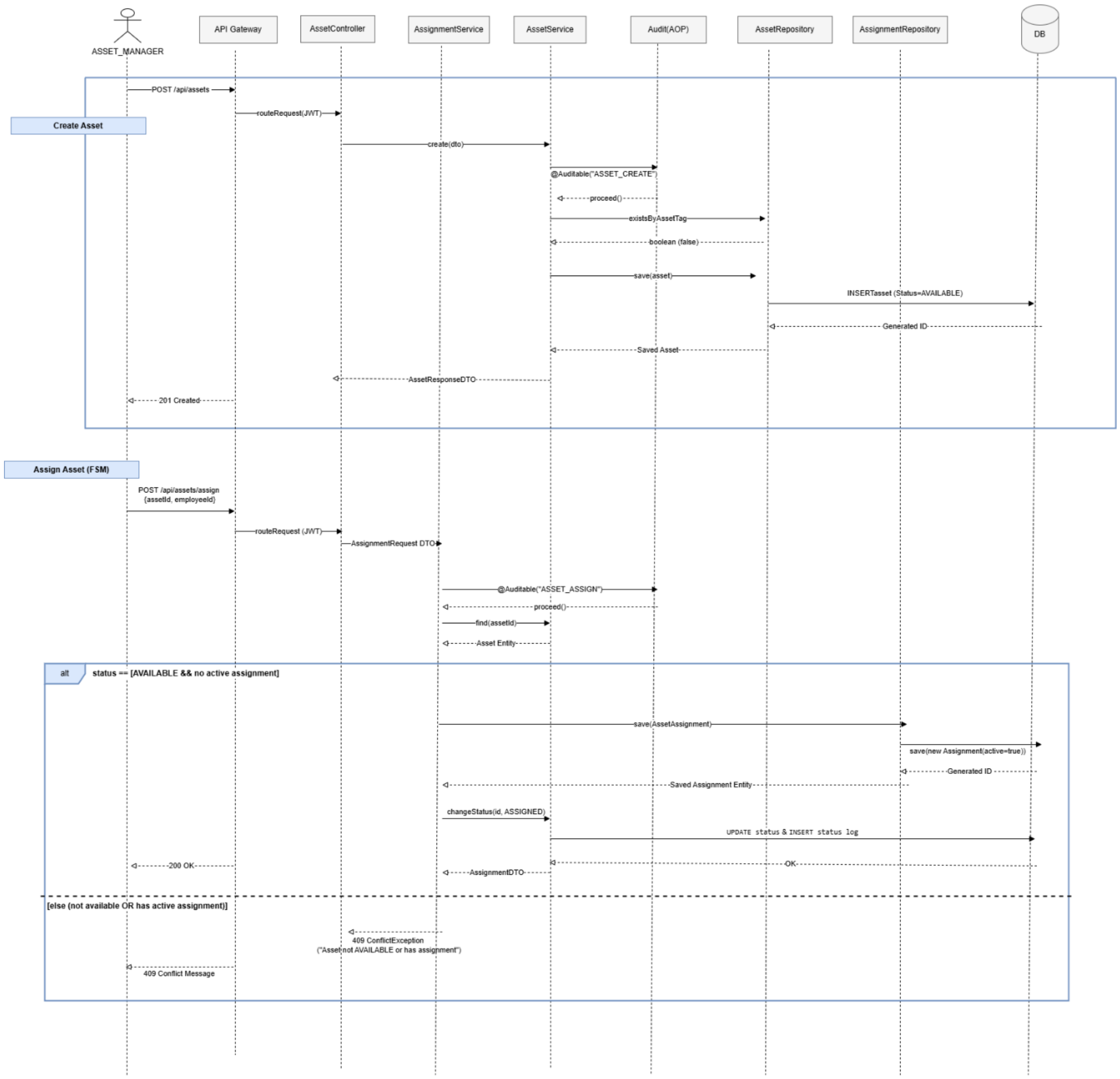
12. Sequence Diagrams

12.1) Maintenance :

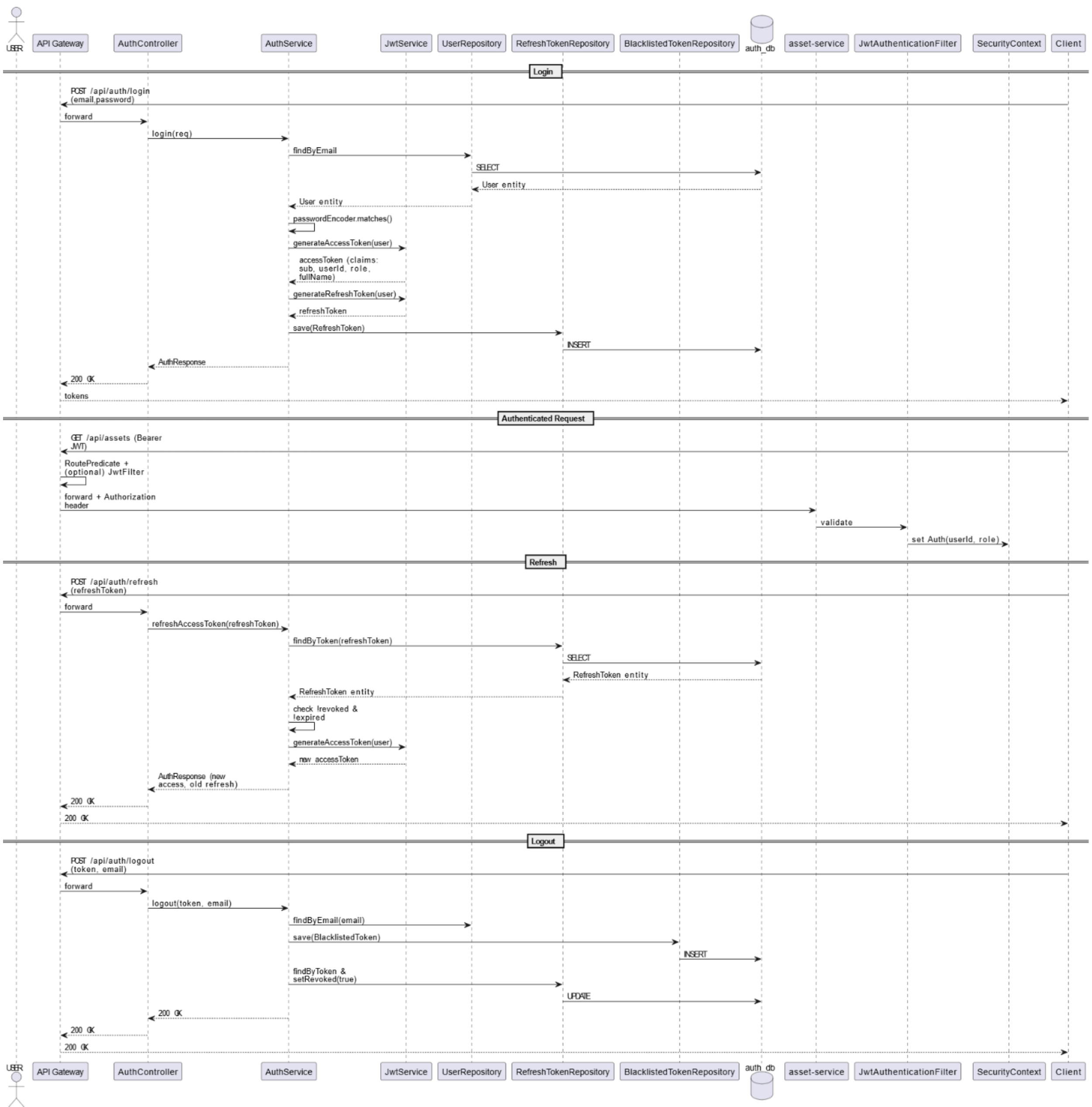
https://drive.google.com/file/d/1bIQVI_I4caLwq_JdNXbo4YOZOagdVAie/view?usp=sharing



12.2) asset sequence : https://drive.google.com/file/d/1exyQJyKm9DTL1_tn4eM1S--j3Ob-QaRB/view?usp=sharing

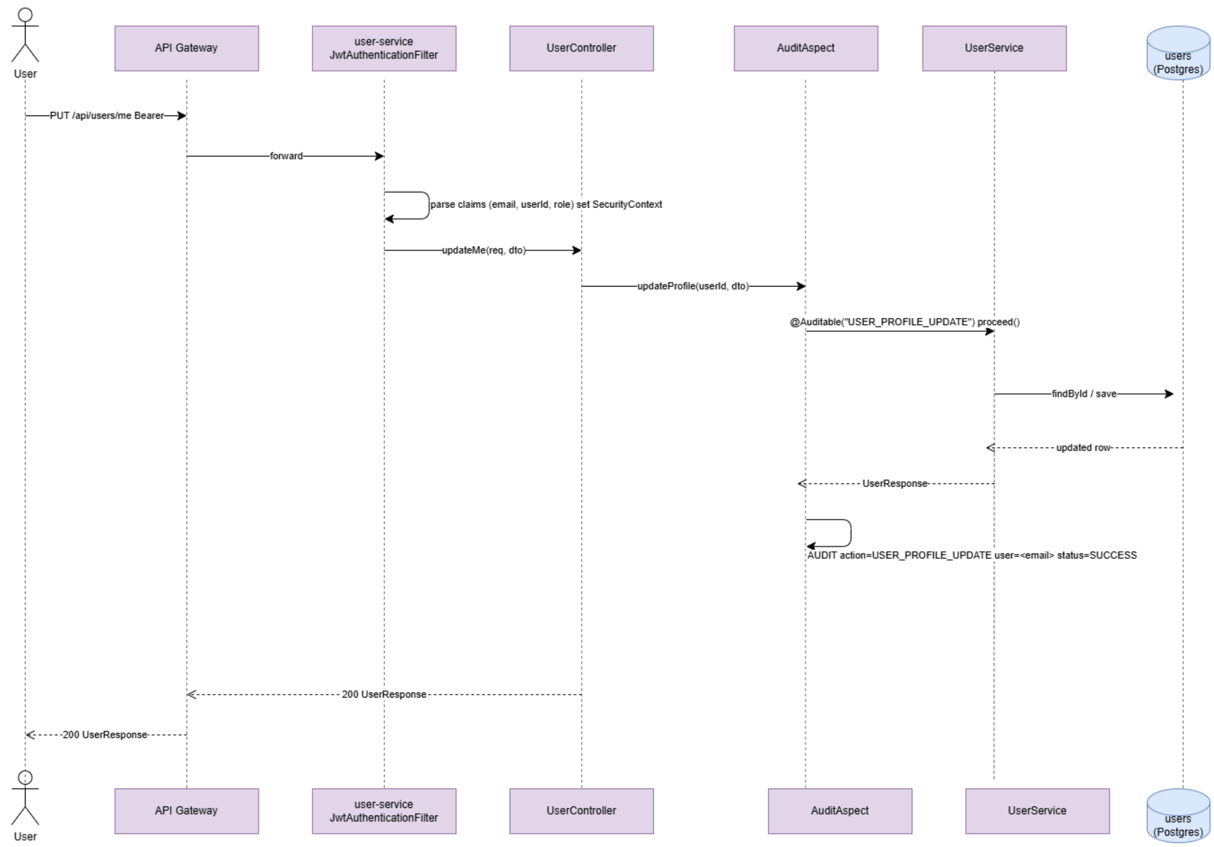


12.3) Authentication sequence :
https://drive.google.com/file/d/1Q0tXx2o6EG5slpaMmhXJ_FIIEVWZUU_7/view?usp=sharing



12.4) user profile sequence : https://drive.google.com/file/d/1j62dLhfkSWknNqAbb9ZW323NLInl6p1J/view?usp=sharing

Sequence — Update Own Profile (user-service)



12.5) report sequence : https://drive.google.com/file/d/1vWaMHmLGavH518TkDQVa8T-FXXz962IP/view?usp=sharing